Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr.S60(N&S), Elite, IAS & LIIT | JEE-MAIN-Shift-1_Batch-II | Date: 17-01-2026

Time: 09.00Am to 12.00Pm | GTM-14 | Max. Marks: 300

KEY SHEET

MATHEMATICS

1	**2**	2	**1**	3	**2**	4	**4**	5	**3**
6	**3**	7	**4**	8	**1**	9	**2**	10	**1**
11	**3**	12	**4**	13	**2**	14	**1**	15	**1**
16	**2**	17	**3**	18	**2**	19	**1**	20	**1**
21	**19**	22	**98**	23	**34**	24	**189**	25	**2**

PHYSICS

26	**1**	27	**2**	28	**1**	29	**3**	30	**1**
31	**1**	32	**2**	33	**4**	34	**1**	35	**1**
36	**1**	37	**3**	38	**2**	39	**4**	40	**2**
41	**3**	42	**3**	43	**3**	44	**3**	45	**1**
46	**16**	47	**121**	48	**3**	49	**36**	50	**30**

CHEMISTRY

51	**2**	52	**2**	53	**1**	54	**2**	55	**4**
56	**2**	57	**2**	58	**1**	59	**1**	60	**3**
61	**1**	62	**2**	63	**4**	64	**2**	65	**2**
66	**2**	67	**3**	68	**4**	69	**2**	70	**2**
71	**1111**	72	**10**	73	**2**	74	**4**	75	**9**

For More Join: @IITJEE_Advanced

SOLUTION
MATHEMATICS

1. $\lim\limits_{x\to 0}\dfrac{f(x)}{x^2}=5$

$\lim\limits_{x\to 0}\dfrac{ax^4+bx^3+cx^2+dx+e)}{x^2}=5$

$c=5$ and $d=e=0$

$f(x)=ax^4+bx^3+5x^2$

$f'(x)=4ax^3+3bx^2+10x$

$=x(4ax^2+3bx+10)$

has extremes at 4 and so f'(4) = 0 & f'(5) = 0

So, $a=\dfrac{1}{8}$ and $b=\dfrac{-3}{2}$

So, $f(2)=\dfrac{1}{8}\times 2^4-\dfrac{3}{2}\times 2^3+5\times 2^2$

$= 2 - 12 + 20 = 10$

2. $P(H)=\dfrac{19}{20}\times\dfrac{1}{2}+\dfrac{1}{20}\times 1$

$\dfrac{19}{20}$: Selection of unbiased

$\dfrac{1}{2}$: Head occurs

$\dfrac{1}{20}$: Selection of biased coin

1: Head occurs

Required probability $=\dfrac{\frac{19}{20}\times\frac{1}{2}}{\frac{19}{20}\times\frac{1}{2}+\frac{1}{20}\times 1}=\dfrac{19}{21}$

$\therefore \dfrac{m}{n}=\dfrac{19}{21}$

$\Rightarrow n^2-m^2=441-361=80$

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$2p+2q=\frac{1}{2}$

$p+q$

$E\left(x^2\right)=\sum_{i=0}^{3}x_i^2p\left(x_i\right)=0\cdot p+1\cdot p+4\cdot q+9q$

$=p+13q$

3. $E\left(x\right)=\sum_{i=0}^{3}x_i^2p\left(x_i\right)=0\cdot p+1\cdot p+2q+3q=p+5q$

$p+13q=2\left(p+5q\right)$

$p=3q$

So, $q=\frac{1}{8}\&p=\frac{3}{8}$

So, $8p-1=2$

$y=\frac{5-x}{x^2-3x+2}$

4. $yx^2-3xy+2y+x-5=0$

$yz^2+\left(-3y+1\right)x+\left(2y-5\right)=0$

Case I : If y = 0 (Accepted)

$\Rightarrow x=5$

Case II : If $y\neq 0$

$D\geq 0$

$(-3y+1)^2-4\left(y\right)\left(2y-5\right)\geq 0$

$9y^2+1-6y-8y^2+20y\geq 0$

$y^2+14y+1\geq 0$

$(y+7)^2-48\geq 0$

$\left|y+7\right|\geq 4\sqrt{3}$

$\Rightarrow y+7\geq 4\sqrt{3}$ or $y+7\leq -4\sqrt{3}$

$\Rightarrow y\geq 4\sqrt{3}-7$ or $y\leq -4\sqrt{3}-7$

From Case I and Case II

$y\in(-\infty,-4\sqrt{3}-7]\cup[4\sqrt{3}-7,\infty)$

So $\alpha=-4\sqrt{3}-7$

$\beta=4\sqrt{3}-7$

$\Rightarrow a^2+b^2=(-4\sqrt{3}-7)^2+(4\sqrt{3}-7)^2$

$=2\left(48+49\right)=194$

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(I) $x < 2$

$-x^2 + 2x - 3x + 9 + 1 = 0$

$\Rightarrow x^2 + x - 10 = 0$

$\Rightarrow x = \frac{-1+\sqrt{41}}{2}, \frac{-1-\sqrt{41}}{2}$

(II) $2 \le x < 3$

$\Rightarrow x^2 - 2x - 3x + 9 + 1 = 0$

$\Rightarrow x^2 - 5x + 10 = 0$

$D < 0$

(III) $x \ge 3$

$x^2 - 2x + 3x - 9 + 2 = 0$

$\Rightarrow x^2 + x - 8 = 0$

$x = \frac{-1+\sqrt{32}}{2}, \frac{-1-\sqrt{32}}{2}$

5. x x

1 real roots

6. $S_n = 700 = \frac{n}{2}\left[2a + (n-1)d\right]$ ...(i)

$a_6 = 7 \Rightarrow a + 5d = 7$ ...(ii)

$S_7 = 7 \Rightarrow \frac{7}{2}(2a + 6d) = 7$

$a + 3d = 1$ ...(iii)

Solve (ii) and (iii)

$\frac{n}{2}(-16 + 3n - 3) = 700 \Rightarrow 3n^2 - 19n - 1400 = 0$

$(3n + 56)(n - 25) = 0$

$\therefore a_{25} = a + 24d = -8 + 24 \times 3$

$= -8 + 72 = 64$

7. $ar + ar^3 + ar^5 = 21, \quad ar^7 + ar^9 + ar^{11} = 15309$

$\Rightarrow ar\left(1 + r^2 + r^4\right) = 21, \quad ar^7\left(1 + r^2 + r^4\right) = 15309$

_____(1) ____(2)

$Eq^n(2) \div eq^n(1)$

$\Rightarrow \frac{a \cdot r^7}{ar} = \frac{15309}{21} \Rightarrow r^6 = 729$

$\Rightarrow \frac{a \cdot (r^9 - 1)}{r - 1} = \frac{\frac{7}{91}(19683 - 1)}{2} = \frac{7 \times 19682}{91 \times 2}$

$= \frac{9841}{13} = 757$

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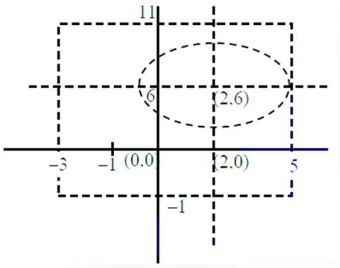
8. $A: |x-1| \le 4$ and $|y-5| \le 6$

$\Rightarrow -4 \le x-1 \le 4 \Rightarrow -6 \le y-5 \le 6$

$\Rightarrow -3 \le x \le 5 \quad \Rightarrow -1 \le y \le 11$

$B: 16(x-2)^2 + 9(y-6)^2 \le 144$

$B: \dfrac{(x-2)^2}{9} + \dfrac{(y-6)^2}{16} \le 1$



From diagram $B \subset A$

9. Length of $LR = \dfrac{2b^2}{a} = 10 \Rightarrow 5a = b^2 \quad ...(1)$

$f(t) = t^2 + t + \dfrac{11}{12}$

$\dfrac{df(t)}{dt} = 2t + 1 = 0 \Rightarrow t = \dfrac{-1}{2}$

Min value of $f(t) = \left(\dfrac{-1}{2}\right)^2 + \left(\dfrac{-1}{2}\right) + \dfrac{11}{12}$

$= \dfrac{1}{4} \dfrac{-1}{2} + \dfrac{11}{12} = \dfrac{3-6+11}{12} = \dfrac{8}{12} = \dfrac{2}{3} = e$

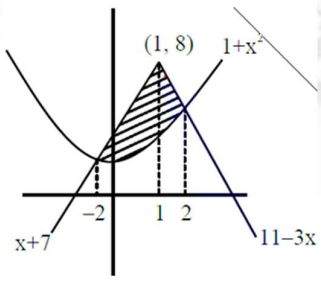
$e^2 = \dfrac{1-b^2}{a^2} \Rightarrow \dfrac{4}{9} = \dfrac{1-b^2}{a^2}$

$\Rightarrow \dfrac{b^2}{a^2} = \dfrac{1-4}{a} = \dfrac{5}{a} \Rightarrow b^2 = \dfrac{5a^2}{a} \ldots\ldots..(2)$

From (1) & (2)

$5a = \dfrac{5a^2}{a} \Rightarrow a = 9, \quad b = \sqrt{45} = 3\sqrt{5}$

$\therefore a^2 + b^2 = 81 + 45 = 126$

10. 

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$$A=\int_{-2}^{1}\left(x+7-x^2-1\right)dx+\int_{1}^{2}\left(11+3x-x^2-1\right)dx$$

$$=\left[\frac{x^2}{2}+6x-\frac{x^3}{3}\right]_{-2}^{1}+\left[10x-\frac{3x^2}{2}-\frac{x^3}{3}\right]_{1}^{2}$$

$$=\frac{50}{3}\Rightarrow 3A=50 \quad \text{Option (1)}$$

$$\text{Re}\left(\frac{z-1}{2z+i}\right)+\text{Re}\left(\frac{\bar{z}-1}{2\bar{z}-i}\right)=2$$

11. $$\text{Here}\frac{z-1}{2z+i}=\left(\frac{\bar{\bar{z}}-1}{2\bar{z}-i}\right)=2$$

$$=\text{Re}\left(\frac{z-1}{2z+i}\right)+\text{Re}\left(\frac{\overline{z-1}}{2z+i}\right)=2$$

$$=2\,\text{Re}\left(\frac{z-1}{2z+1}\right)=2\Rightarrow\text{Re}\left(\frac{z-1}{2z+i}\right)=1$$

Let z = x + iy

$$\text{Re}\left(\frac{(x-1)+iy}{2x+i(2y+1)}\right)=1\Rightarrow\text{Re}\left[\frac{((x-1)+iy)(2x-i(y+1)}{(2x+i(2y+1)(2x-i(2y+1))}\right]=1$$

$$\Rightarrow\frac{2x(x-1)+y(2y+1)}{4x^2+(2y+1)^2}=1$$

$$\Rightarrow 2x^2-2x+2y^2+y=4x^2+4y^2+1+4y$$

$$\Rightarrow 2x^2+2y^2+3y+2x+1=0$$

$$\Rightarrow x^2+y^2+x+\frac{3}{2}y+\frac{1}{2}=0$$

$$\text{centre}=\left(\frac{-1}{2},\frac{-3}{4}\right),r=\sqrt{\frac{1}{4}+\frac{9}{16}-\frac{1}{2}}=\frac{\sqrt{5}}{4}$$

$$a=\frac{-1}{2},b=\frac{-3}{4},r^2=\frac{5}{16}$$

$$15\frac{ab}{r^2}=15\times\left(\frac{-1}{2}\right)\times\left(\frac{-3}{4}\right)\times\frac{16}{5}=18$$

12. Total triangles $\Rightarrow={}^{h}C_3$

Total audrilaterals $={}^{h}C_4=q$

$${}^{n}C_3+{}^{n}C_4=126\Rightarrow{}^{n+1}C_4=126$$

$$\Rightarrow n+1=9\Rightarrow n=8$$

$$\frac{x^2}{16}+\frac{y^2}{n}=1\Rightarrow\frac{x^2}{16}+\frac{y^2}{8}=1$$

$$e=\sqrt{1-\frac{8}{16}}=\sqrt{\frac{8}{16}}=\frac{1}{\sqrt{2}}$$

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$e_1^2 = 1 - \frac{b^2}{25}$ $e_2^2 = 1 - \frac{b^2}{16}$

$\therefore e_1^2 e_2^2 = 1$

$\left(1 - \frac{b^2}{25}\right)\left(1 + \frac{b^2}{16}\right) = 1$

$\Rightarrow 2 + \frac{b^2}{16} - \frac{b^2}{25} - \frac{b^2}{400} = 1$

$\Rightarrow \frac{9b^2}{400} = \frac{b^4}{400}$

$b^2 = 9$

$\frac{x^2}{9} + \frac{y^2}{25} = 1$ $\frac{x^2}{16} - \frac{y^2}{9} = 0$

$e_1 \sqrt{1 - \frac{9}{25}}$

13. $e_1 = \frac{4}{5}$

Focii : (0,±4) (±5,0)

ellipse passing through all four foci

$\frac{x^2}{25} + \frac{y^2}{16} = 1$

$e = \sqrt{1 - \frac{16}{25}} = \frac{3}{5}$

14. For infinitely many solution

$\Delta = 0$

$$\begin{vmatrix} 1 & 5 & -1 \\ 4 & 3 & -3 \\ 24 & 1 & \lambda \end{vmatrix} = 0$$

$\Rightarrow 1(3\lambda + 3) - 5(4\lambda + 72) - 1(4 - 72) = 0$

$\Rightarrow -17\lambda + 3 - 4 \times 72 - 4 = 0$

$\Rightarrow 17\lambda = -289$

$\Rightarrow \lambda = -17$

$\Delta 1 = 0$

$$\Rightarrow \begin{vmatrix} 1 & 5 & -1 \\ 7 & 3 & -3 \\ \mu & 1 & -17 \end{vmatrix} = 0$$

$\Rightarrow 1(-51 + 3) - 5(-119 + 3\mu) - 1(7 - 3\mu) = 0$

$\Rightarrow -48 + 595 - 15\mu - 7 + 3\mu = 0$

$\Rightarrow 12\mu = 540$

$\mu = 45$

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$x+5y-z=1$

$4x+3y-3z=7$

$24x+y-17z=45$

Let $z=1$

$x+5y=1+\lambda]\times 4$

$4x+3y=7+3\lambda$

$4x+20y=-4+4\lambda$

$\frac{-17y=3-\lambda}{-17}$

$y=\frac{\lambda-3}{17}, x=1+\lambda-\frac{5\lambda-15}{17}$

$=\frac{32-12\lambda}{17}$

$7\le\frac{\lambda-3}{17}+\frac{32+12\lambda}{17}+\lambda\le 77$

$7\le\frac{30\lambda+29}{17}\le 77$

$3\le\lambda\le 42$

$\lambda=3,20,37$

15. $\left(x^2+1\right)\frac{dy}{dx}-2xy=\left(x^4+2x^2+1\right)\cos x$

$\frac{dy}{dx}-\left(\frac{2x}{x^2+1}\right)y=\frac{\left(x^2+1\right)^2\cos x}{cx^2+1}=\left(x^2+1\right)\cos x$

(Linear D.E)

$P=\frac{-2x}{x^2+1}, Q=\left(x^2+1\right)\cos x$

$I.F=e^{\int Pdx}=e^{\int\frac{-2x}{x^2+1}dx}=\frac{1}{x^2+1}$

$y\cdot\frac{1}{x^2+1}=\int\left(x^2+1\right)\cos x\cdot\frac{1}{x^2+1}dx$

$\frac{y}{x^2+1}=\sin x+c\Rightarrow y\cos=1\Rightarrow c=1$

$y=\left(x^2+1\right)(\sin x+1)$

$\int_{-3}^{3}ydx=\int_{-3}^{3}\left(x^2+1\right)(\sin x+1)$

$dx=\int_{-3}^{3}x^2\sin x+x^2\sin x+1dx$

$\Rightarrow\int_{-3}^{3}x^2\sin xdx+\int_{-3}^{3}x^2dx+\int_{-3}^{3}\sin xdx+\int_{-3}^{3}1dx$

$=0+18+0+6=24$

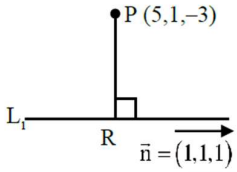
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$L_1 : \frac{x-1}{1} = \frac{y-2}{1} = \frac{z-0}{2}$

16. Let $Q(\lambda+1, \lambda+2, \lambda)$

$\overrightarrow{PQ} = (\lambda-4, \lambda-1, \lambda+3)$

$\overrightarrow{PQ} \cdot \vec{m} = 0$



$\Rightarrow \lambda - 4 + \lambda + 1, \lambda + 3 = 0$

$\Rightarrow 3\lambda = 0$

$\Rightarrow Q(1,2,0)$

$L_2 : \frac{x-2}{1} = \frac{y-0}{1} = \frac{z-1}{2}$

Let $R(\mu+2, \mu, \mu+1)\ \overrightarrow{PR} = (\mu-3, \mu-1, \mu+4)$

$\overrightarrow{PR} \cdot \vec{n} = 0$

$\mu - 3 + \mu - 1 + \mu + 4 = 0$

$\neq \mu = 0$

$R(2,0,1)$

Area of $\Delta PQR(A) = \frac{1}{2}\left|\overrightarrow{PQ} \times \overrightarrow{PR}\right|$

$A = \frac{1}{2}\left|\left(-4\hat{i} + \hat{j} + 3\hat{k}\right) \times \left(-3\hat{i} + \hat{j} + 4\hat{k}\right)\right|$

$A = \frac{1}{2}\left|7\left(\hat{i} + \hat{j} + \hat{k}\right)\right|$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -4 & 1 & 3 \\ -3 & -1 & 4 \end{vmatrix}$$

$= 7\hat{i} + 7\hat{j} + 7\hat{k}$

$4A^2 = 49 \times 3 = 147$

17. $\frac{|\bar{a}+\bar{b}|+|\bar{a}-\bar{b}|}{|\bar{a}+\bar{b}|-|\bar{a}-\bar{b}|} = \sqrt{2}+1$

Apply componendo and dividend

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$$\Rightarrow \frac{2\left|\bar{a}+\bar{b}\right|}{2\left|\bar{a}-\bar{b}\right|}=\frac{\sqrt{2}+2}{\sqrt{2}}$$

$$\Rightarrow \left|\bar{a}+\bar{b}\right|=\left(1+\sqrt{2}\right)\left|\bar{a}-\bar{b}\right|$$

$$\Rightarrow \left|\bar{a}+\bar{b}\right|^2=\left(3+2\sqrt{2}\right)\left|\bar{a}-\bar{b}\right|^2$$

$$\Rightarrow 2|\bar{a}|^2+2\bar{a}\cdot\bar{b}=\left(3+2\sqrt{2}\right)\left(2|\bar{a}|^2-2\bar{a}\cdot\bar{b}\right)$$

$$\Rightarrow 2|\bar{a}|^2\left(2+2\sqrt{2}\right)=2\bar{a}\cdot\bar{b}\left(4+2\sqrt{2}\right)$$

$$\Rightarrow \frac{\bar{a}\cdot\bar{b}}{|\bar{a}|^2}=\frac{2+2\sqrt{2}}{4+2\sqrt{2}}=\frac{1}{\sqrt{2}}$$

$$\text{Now, } \frac{|\bar{a}+\bar{b}|^2}{|\bar{a}|^2}=1+\frac{|\bar{b}|^2}{|\bar{a}|^2}+\frac{2\bar{a}\cdot\bar{b}}{|\bar{a}|^2}$$

$$=1+1+2\left(\frac{1}{\sqrt{2}}\right)=2+\sqrt{2}$$

18. Line is $\perp^r$ to2 line $\Rightarrow$line will be parallel to $\left(\hat{i}+a\hat{j}+b\hat{k}\right)\times\left(-b\hat{i}+a\hat{j}+5\hat{k}\right)$

Parallel vector along the required line is $\hat{i}\left(5a-ab\right)-\hat{j}\left(b^2+5\right)+\hat{k}\left(a+ab\right)$

Dr's of required line $\alpha\left(5a-ab\right),-\left(b^2+5\right),\left(a+ab\right)$

Also Dr's of the required line α-2,d,-4

$$\therefore \frac{5a-ab}{-2}=\frac{-\left(b^2+5\right)}{d}=\frac{a+ab}{-4}\ldots(1)$$

Also point, $\left(0,\frac{-1}{2},0\right)$ will lie on $\frac{x-1}{-2}=\frac{y+4}{d}=\frac{z-c}{-4}$

$$\frac{0-1}{-2}=\frac{\frac{-1}{2}+4}{d}=\frac{0-c}{-4}\Rightarrow d=7,c=2$$

$$\text{From(1)}\frac{5a-ab}{-2}=\frac{-b^2-5}{7}=\frac{a+ab}{-4}$$

$$\frac{5a-ab}{-2}=\frac{a+ab}{-4};\frac{-b^2-5}{7}=\frac{a+ab}{-4}$$

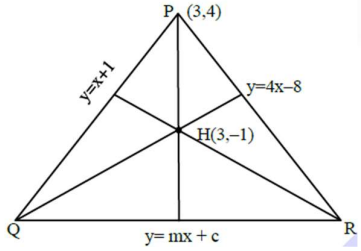
$$-20a+4ab=-2a-2ab \qquad 4b^2+20=70+7ab$$

$$18a=6ab \qquad 36+20=70+21a$$

$$b=3 \qquad 56=28a\Rightarrow a=2$$

$$a+b+c+d=2+3+2+7=14$$

19.



Solve line PQ & QR

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Point $Q\left(\frac{1-c}{m-1}, \frac{1-c}{m-1}+1\right)$

$$m_{2H}=\frac{\frac{1-c}{m-1}+2}{\frac{1-c}{m-1}-3}=\frac{1-c+2m-2}{1-c-3m+3}=-\frac{1}{4} \quad ...(1)$$

$\because m_{PH}=\frac{5}{0}\to\infty$

$\Rightarrow$Slope of line QR(m)=0

Put value of m in equation (1)

$$\frac{1-c-2}{1-c+3}=-\frac{1}{4}\Rightarrow c=0$$

$$\cos 2\theta\cos\frac{\theta}{2}+\cos\frac{5\theta}{2}=2\cos^3\frac{5\theta}{2}$$

20. $$\frac{1}{2}\left(2\cos 2\theta\cos\frac{\theta}{2}\right)+\cos\frac{50}{2}$$

$$=\frac{1}{2}\left(\cos\frac{15\theta}{2}+3\cos\frac{5\theta}{2}\right)$$

Or solving

$$\cos\frac{3\theta}{2}=\cos\frac{15\theta}{2}$$

$$\cos\frac{15\theta}{2}-\cos\frac{3\theta}{2}=0$$

$$2\sin 30\sin\frac{9\theta}{2}=0$$

$$3\theta=n\pi \text{ or } \frac{9\theta}{2}=m\pi$$

$$\theta=\frac{n\pi}{3} \quad \theta=\frac{2m\pi}{9}$$

$$\theta=\left\{-\frac{\pi}{2},\frac{\pi}{3},0\right\}$$

$$\theta=\left\{-\frac{4\pi}{9},\frac{-2\pi}{9},\frac{4\pi}{9},\frac{2\pi}{9}\right\}$$

$$\int\left(\frac{1}{x^2}+\frac{1}{x^4}\right)\left(\frac{3}{x}+\frac{1}{x^3}\right)^{\frac{1}{23}}dx$$

21. $$\text{Using } t=\frac{3}{x}+\frac{1}{x^3}\Rightarrow dt=-3\left(\frac{1}{x^2}+\frac{1}{x^4}\right)dx$$

$$\int\frac{t^{1/23}dt}{-3}=\frac{t^{24/23}}{\left(\frac{24}{23}\right)(-3)}+C$$

$$\Rightarrow \alpha=23,\beta=-1,\gamma=-3$$

$$\therefore \alpha+\beta+\gamma=19$$

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22. $a+b=\lim\limits_{t\to-1^+}(\alpha+\beta)=\lim\limits_{t\to-1^+}-\dfrac{(t+2)^{\frac{1}{6}}-1}{(t+2)^{\frac{1}{7}}-1}$

let $t+2=y$

$a+b=\lim\limits_{y\to1^+}\dfrac{y^{1/6}-1}{y^{1/7}-1}=\dfrac{7}{6}$

$72(a+b)^2=72\dfrac{49}{36}=98$

$(1-x)^{20}={}^{20}C_0-{}^{20}C_1x+{}^{20}C_2x^2\ldots.+{}^{20}C_{20}x^{20}$

23. $\dfrac{(1-x)^{20}}{x^2}=\dfrac{{}^{20}C_0}{x^2}-\dfrac{{}^{20}C_1}{x}+{}^{20}C_2-{}^{20}C_3x+{}^{20}C_4x^2\ldots$

Diff twice and put $x=1$

$=6-{}^{20}C_1(2)+A$

$A=40-6=34$

$\left.\begin{matrix}ae=5\\ \dfrac{a}{e}=9/5\end{matrix}\right\}\begin{matrix}a=3\\ e=5/3\\ b=4\end{matrix}$

24. $\dfrac{x^2}{9}-\dfrac{y^2}{16}=1$

$\dfrac{\alpha^2}{9}-\dfrac{4(5)}{16}=1\Rightarrow\alpha^2=9\left(\dfrac{36}{16}\right)$

$PF_1\cdot PF_2=(e\alpha-a)(e\alpha+a)$

$p=e^2\alpha^2-a^2=\dfrac{25}{9}\cdot9\cdot\dfrac{9}{4}-9=\dfrac{189}{4}$

25. $\lim\limits_{x\to0}\dfrac{\dfrac{\tan(\tan x)-\tan x}{\tan^3x}\dfrac{\tan^3x}{x^3}+\dfrac{\tan x-\sin x}{x^3}+\dfrac{\sin x-\sin(\sin x)}{\sin^3x}\dfrac{\sin^3x}{x^3}}{\dfrac{\tan x-\sin x}{x^3}}$

$=\dfrac{\dfrac{1}{3}+\dfrac{1}{2}+\dfrac{1}{6}}{\dfrac{1}{2}}=2$

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PHYSICS

$PV = nRT$

$PV = \frac{N}{N_A}RT$

N = Total no. of molecules

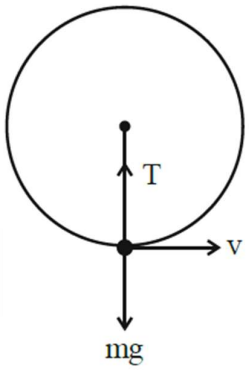
26. $P = \frac{N}{V}kT$

$1.38\times1.01\times10^{5} = 2\times10^{25}\times1.38\times10^{-23}\times T$

$1.01\times10^{5} = 2\times10^{2}\times T$

$T = \frac{1.01\times10^{3}}{2} \approx 500K$

27. Given that



$m = 900gm = \frac{900}{1000}kg = \frac{9}{10}kg$

$r = 1m$

$\omega = \frac{2\pi N}{60} = \frac{2\pi(10)}{60} = \frac{\pi}{3}rad/sec$

$T - mg = mr\omega^2$

$T = mg + mr\omega^2$

$= \frac{9}{10}\times9.8 + \frac{9}{10}\times1\left(\frac{\pi}{3}\right)^2$

$= 8.82 + \frac{9}{10}\times\frac{\pi^2}{9}$

$= 8.82 + 0.98$

$= 9.80N$

$\ell = 10m$

Initial energy $= mg\ell$

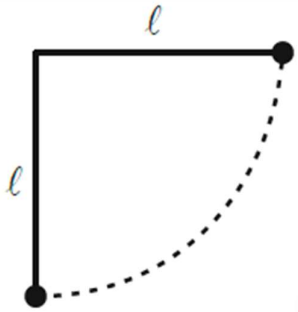
28. So, $\frac{9}{10}mg\ell = \frac{1}{2}mv^2$

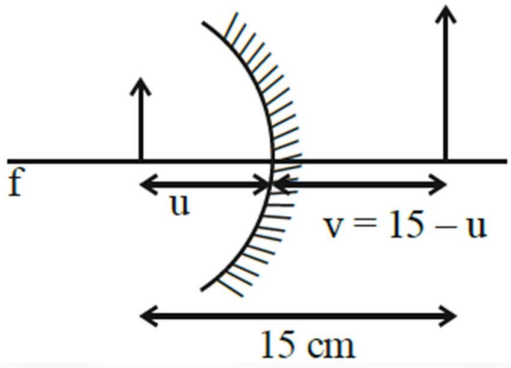
$\Rightarrow \frac{9}{10}\times10\times10 = \frac{1}{2}v^2$

$v^2 = 180$

$v = \sqrt{180} = 6\sqrt{5}m/s$

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29.

$$m = 2 = \frac{-v}{u}$$

$$2 = \frac{-(15-u)}{-u}$$

$$2u = 15 - u$$

$$3u = 15 \Rightarrow u = 5\,cm$$

$$v = 15 - u = 15 - 5 = 10\,cm$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$= \frac{1}{10} + \frac{1}{(-5)} = \frac{1-2}{10} = \frac{-1}{10}$$

$$f = -10\,cm$$

$$\Delta x = \frac{7\lambda}{4}$$

$$\phi = \frac{2\pi}{\lambda}\Delta x = \frac{2\pi}{\lambda} \times \frac{7\lambda}{4} = \frac{7\pi}{2}$$

$$I = I_{max}\cos^2\left(\frac{\phi}{2}\right)$$

$$\frac{I}{I_{max}} = \cos^2\left(\frac{\phi}{2}\right) = \cos^2\left(\frac{7\pi}{2\times 2}\right) = \cos^2\left(\frac{7\pi}{4}\right)$$

$$= \cos^2\left(2\pi - \frac{\pi}{4}\right)$$

30.

$$= \cos^2\frac{\pi}{4}$$

$$= \frac{1}{2}$$

31. Volume constant

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$$\frac{4}{3}\pi R^3 = 27\times\frac{4}{3}\times\pi r^3$$

$$R^3 = 27r^3$$

$$R = 3r$$

$$r = \frac{R}{3}$$

$$r^2 = \frac{R^2}{9}$$

Work done $= T.\Delta A$

$$= 27T\left(4\pi r^2\right) - T4\pi R^2$$

$$= 27T4\pi\frac{R^2}{9} - 4\pi R^2 T$$

$$= 8\pi R^2 T$$

32. Apply energy conservation between A & B

$$\frac{1}{2}mV_L^2 = \frac{1}{2}mV_H^2 + mg(2L)$$

$$\because V_L = \sqrt{5gL}$$

$$\text{So}, V_H = \sqrt{gL}$$

$$\frac{(K.E)_A}{(K.E)_B} = \frac{\frac{1}{2}m(\sqrt{5gL})^2}{\frac{1}{2}m(\sqrt{gL})^2} = \frac{5}{1}$$

$$Y = \frac{\text{stress}}{\text{strain}}$$

$$Y = \frac{\frac{F}{\pi r^2}}{\frac{\ell}{L}}$$

$$F = Y\pi r^2\times\frac{\ell}{L}$$

33.

$$Y = \frac{F/2}{\frac{\pi r^2/4}{L}}$$

$$F = Y\frac{\Delta\ell}{L}\times 2\times\frac{\pi r^2}{4}$$

From (i)

$$Y\pi r^2\frac{\ell}{L} = Y\frac{\Delta\ell}{L}\frac{\pi r^2}{2}$$

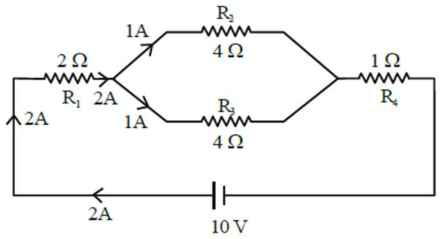
$$\Delta\ell = 2\ell$$

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34.
$$T^2 \propto r^3$$
$$\frac{T_1^2}{r_1^3} = \frac{T_2^2}{r_2^3}$$
$$\frac{(200)^2}{r^3} = \frac{T_2^2}{\left(\frac{r}{4}\right)^3}$$
$$\frac{200\times 200}{4\times 4\times 4} = T_2^2$$
$$T_2 = \frac{200}{4\times 2}$$
$$T_2 = 25\text{days}$$

35.
$$\frac{E}{B} = C$$
$$\frac{E}{B} = 3\times 10^8$$
$$B = \frac{E}{3\times 10^8} = \frac{9.6}{3\times 10^8}$$
$$B = 3.2\times 10^{-8}\text{T}$$
$$\hat{B} = \hat{v}\times\hat{E}$$
$$\text{So, } \vec{B} = 3.2\times 10^{-8}\widehat{kT}$$

36.



$$R_{eq} = 2\Omega + 2\Omega + 1\Omega = 5\Omega$$
$$i = \frac{V}{R_{eq}} = \frac{10}{5} = 2\text{A}$$
$$\text{Current in resistance } R_3 = 2\times\left(\frac{4}{4+4}\right)$$
$$= 2\times\frac{4}{8}$$
$$= 1\text{A}$$

37.
$$\vec{E} = 6\hat{i} + 5\hat{j} + 3\hat{k}$$
$$\vec{A} = 30\hat{i}$$
$$\phi = \vec{E}\cdot\vec{A}$$
$$\phi = \left(6\hat{i} + 5\hat{j} + 3\hat{k}\right)\cdot\left(30\hat{i}\right)$$
$$\phi = 6\times 30 = 180$$

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$$R=\frac{mv}{qB}=\frac{p}{qB}=\frac{\sqrt{2m(KE)}}{qB}=\frac{\sqrt{2mqV}}{qB}$$

38. $R\propto\sqrt{m}$

$m\propto R^2$

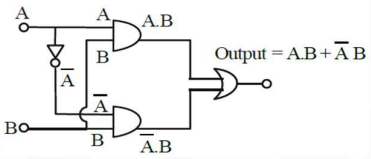
$$\frac{m_1}{m_2}=\left(\frac{R_1}{R_2}\right)^2$$

39. $$n_1\times\frac{hc}{\lambda_1}=200$$

$$n_2\times\frac{hc}{\lambda_2}=200$$

$$\frac{n_1}{n_2}=\frac{\lambda_1}{\lambda_2}=\frac{300}{500}$$

$$\frac{n_1}{n_2}=\frac{3}{5}$$

40. 

$Y=A\cdot B+\bar{A}\cdot B$

$=(A+\bar{A})\cdot B$

$Y=1\cdot B\Rightarrow Y=B$

$$Q=\frac{a^4b^3}{c^2}$$

41. $$\frac{\Delta Q}{Q}=4\frac{\Delta a}{a}+3\frac{\Delta b}{b}+2\frac{\Delta c}{c}$$

$$\frac{\Delta Q}{Q}\times100=4\left(\frac{\Delta a}{a}\times100\right)+3\left(\frac{\Delta b}{b}\times100\right)+2\left(\frac{\Delta c}{c}\times100\right)$$

$\%\text{error in}Q=4\times3\%+3\times4\%+2\times5\%$

$=12\%+12\%+10\%$

$=34\%$

$$P_{avg}=V_{rms}I_{rms}\cos(\Delta\phi)$$

42. $$=\frac{100}{\sqrt{2}}\times\frac{100\times10^{-3}}{\sqrt{2}}\times\cos\left(\frac{\pi}{3}\right)$$

$$=\frac{10^4}{2}\times\frac{1}{2}\times10^{-3}$$

$$=\frac{10}{4}=2.5\text{W}$$

43. According to Rutherford atomic model, most of mass of atom and all its positive charge is concentrated in tiny nucleus & electron revolve around it.
According to Thomson atomic model, atom is spherical cloud of positive charge with electron embedded in it.
Hence, Statement I is true but statement II false.

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$x = t^3 - 6t^2 + 20t + 15$

$\frac{dx}{dt} = v = 3t^2 - 12t + 20$

$\frac{dv}{dt} = a = 6t - 12$

44. When $a = 0$

$6t - 12 = 0; t = 2\,sec$

$Ats = 2\,sec$

$v = 3(2)^2 - 12(2) + 20$

$v = 8m/s$

45. $f_{eq} = \frac{n_1 f_1 + n_2 f_2}{n_1 + n_2}$

For diatomic gas $f_{eq}=5$

$5 = \frac{(N)(6)+(2)(3)}{N+2}$

$5N + 10 = 6N + 6$

$N = 4$

46.

$Y = \frac{Stress}{Strain} = \frac{F/A}{\Delta\ell/\ell} = \frac{F\ell}{A\Delta\ell}$

$\Delta\ell = \frac{F\ell}{AY}$

$V = A\ell \Rightarrow \ell = \frac{V}{A}$

$\Delta\ell = \frac{FV}{A^2Y}$

Y & V is same for both the wires

$\Delta\ell \propto \frac{F}{A^2}$

$\frac{\Delta\ell_1}{\Delta\ell_2} = \frac{F_1}{A_1^2} \times \frac{A_2^2}{F_2}$

$\Delta\ell_1 = \Delta\ell_2$

$F_1A_2^2 = F_2A_1^2$

$\frac{F_1}{F_2} = \frac{A_1^2}{A_2^2} = \left(\frac{4}{1}\right)^2 = 16$

47. For minimum potential difference electron has to make transition from n = 3 to n = 2 state but first electron has to reach to n = 3 state from ground state. So, energy of bombarding electron should be equal to energy difference of n = 3 and n = 1 state.

$\Delta E = 13.6\left[1 - \frac{1}{3^2}\right]e = eV$

$\frac{13.6 \times 8}{9} = V$

$V = 12.09V \approx 12.1V$

$So, \alpha = 121$

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48.

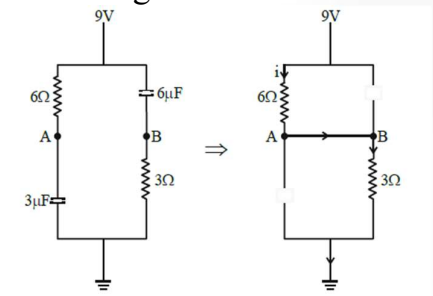
$B_H = 0.60\times10^{-4}\,Wb/m^2$

Induced emfe $= B_H v\ell$

$= 0.60\times10^{-4}\times10\times5$

$= 3\times10^{-3}\,V$

49. At steady state, capacitor behaves as an open circuit and current flows in circuit as shown in the diagram.



$R_{eq} = 9\Omega$

$i = \dfrac{9V}{9\Omega} = 1A$

$\Delta V_{6\Omega} = 1\times6 = 6V$

$V_A = 3V$

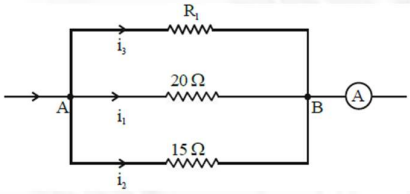
So, potential difference across 6μF is 6 V.

Hence Q=CΔV

$= 6\times6\times10^{-6}\,C$

$= 36\mu C$

50.



Given, $i_1 = 0.3A, i_1 + i_2 + i_3 = 0.9A$

So, $V_{AB} = i_1\times20\Omega = 20\times0.3V = 6V$

$i_2 = \dfrac{6V}{15\Omega} = \dfrac{2}{5}A$

$i_1 + i_2 + i_3 = \dfrac{9}{10}A$

$\dfrac{3}{10} + \dfrac{2}{5} + i_3 = \dfrac{9}{10}$

$\dfrac{7}{10} + i_3 = \dfrac{9}{10}$

$i_3 = 0.2A$

So, $i_3\times R_1 = 6V$

$(0.2)R_1 = 6$

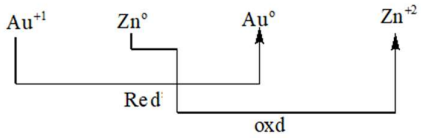
$R_1 = \dfrac{6}{0.2} = 30\Omega$

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CHEMISTRY

$$2\left[Au(CN)_2\right]^-(aq)+Zn(s)\rightarrow 2Au(s)+\left[Zn(CN)_4\right]^{2-}(aq)$$

51.



$$2\left[Au(CN)_2\right]^-(aq)+Zn(s)\rightarrow 2Au(s)+\left[Zn(CN)_4\right]^{2-}(aq)$$

Au^{+1} Zn° Au° Zn^{+2}

It is a redox reaction

Zn displaces Au from complex Statements A & B are correct.

52. A) Hell – Volhard – Zelinsky reaction

i) Br_2/ red phosphorus

ii) H_2O

B) Idoform Reaction − $NaOH+I_2$

C) Etard Reaction - (i) CrO_2Cl_2,CS_2

(ii) H_2O

D) Gattermann - Koch reaction – CO, HCl, anhyd $AlCl_3$

53. The standard electrode potential of M^+/M in aqueous solution does not depend upon ionization of a solid metal atom.

54. When an electron loses a hydrogen atom, the nucleus (H^+) shrinks to 1.5×10^{-3}pm in size, which is extremely small in comparison to typical atomic or ionic sizes.

Hence, both A and R are correct but R is NOT the correct explanation of A

55.

$$r=\frac{0.529\times n^2}{2}$$

$$r_1=\frac{a_0\times(1)^2}{1}=a_0$$

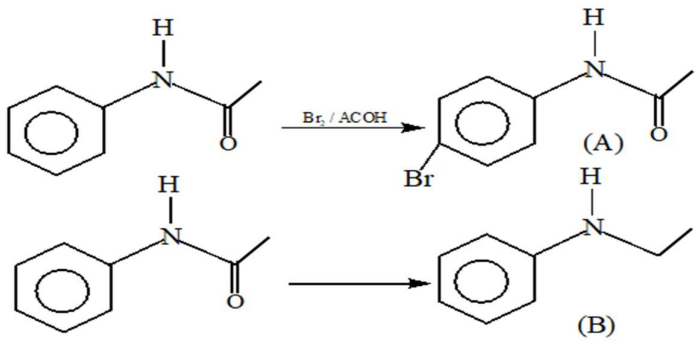
$$r_3=\frac{a_0\times3^2}{1}=9a_0$$

$$2\pi r_3=n\lambda$$

$$2\pi 9a_0=3\lambda$$

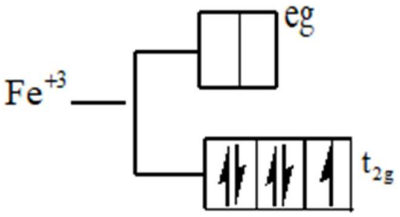
$$6\pi a_0=\lambda$$

56.



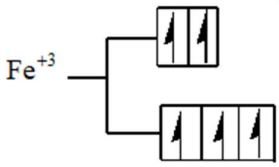
57. Assertion At $\left[Fe(CN)_6\right]^{3-}$ CN is a strong field ligand. Therefore pairing occurs first

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no. of unpaired e⁻s=1

H_2O is a weak field ligand. Therefore splitting occurs.



No of unpaired $e^{(-)}S=1$ $\therefore$ Magnetic moment $=\sqrt{5(7)}=5.92$ BM.

Reason R : In both complexes, Fe is present in +3 oxidation state. But it does not explain assertion.

$\therefore$Both A and R are true but R is not correct explanation of A.

58. Number intermediates → bottom points of the 2 curves
Number of activated complexes → 3peaks
Rate determining step → large peak II points 3

59. Eu exhibits both +2 and +3 oxidation states similarly, Ce exhibits +3 and +4 oxidation states.
Based on E^o value, Ce^{+4} acts as strong oxidant.
However, Eu^{+2} is a strong reductant as it changing to the common +3 state.

60. Vitamin A – Xeropthalmia
Thiamine - Beri – Beri
Ascorbic acid – Scurvy
Riboflavin – Cheilosis

61. (A)Transition state is less polar than the localized anion (correct)
(B) The solvent taken is polar aprotic solvent so solvation cannot occur
(C) No, Br^- cannot act as competing nucleophile as it is precipitated in acetone
(D) No, the above reaction cannot occur in acetic acid.

62. $A_2B_3 \rightleftharpoons 2A^{3+} + 3B^{2-}$
Initial C
final $C-C\alpha$ $2C\alpha$ $(3C\alpha)$
From law of mass action

$$K_{eq}=\frac{(2C\alpha)^2\times(3C\alpha)^3}{(C-C\alpha)}$$

$$\alpha<<1\Rightarrow 1-\alpha\simeq 1$$

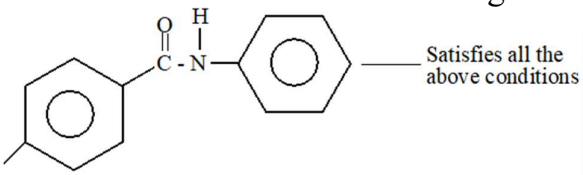
$$K_{eq}=\frac{2^2\times3^3\times C^5\times\alpha^5}{(1-\alpha)\times C}\Rightarrow K_{eq}=108C^4\times\alpha^5$$

$$\Rightarrow\alpha=\left(\frac{K_{eq}}{108C^4}\right)^{\frac{1}{5}}$$

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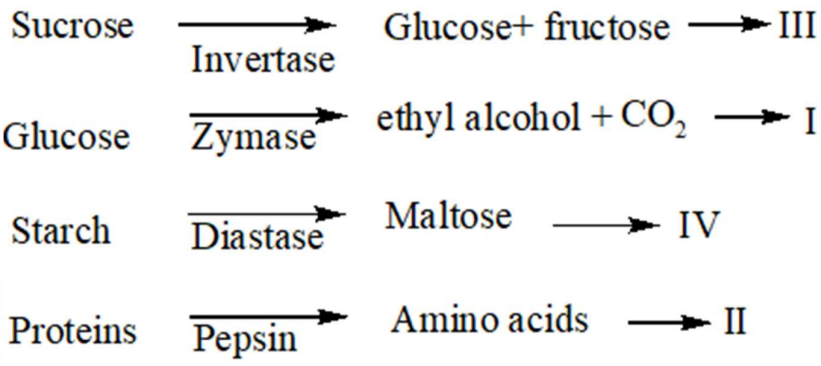
63. In the periodic table 'Cl' has most negative electron gain enthalpy and neon has the highest positive electron gain enthalpy
∴ difference is highest for Ne and Cl

64. Compound P undergoes hydrolysis an addition of HCl residue Q should be a carboxylic acid and R should be a amine as it gives soluble solid in NaOH in Hinsberg test.



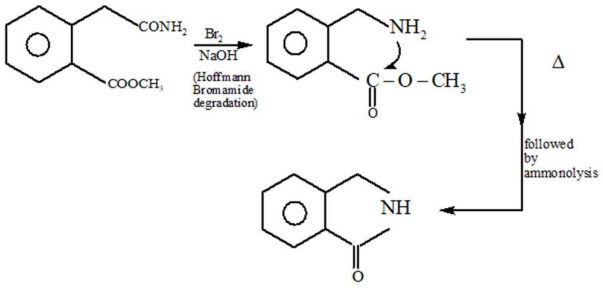
65. The component in mobile phase have more polar is reactive absorbed Here a is reactive adsorbed and strong interaction between mobile phase & stationary phase.

66.



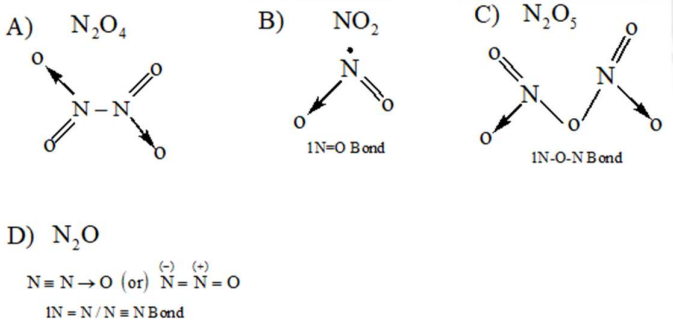
67.
Nitrogen – $Fe_4[Fe(CN)_6]_3$
Sulphur – $Na_2[Fe(CN)_5NO]$
Phosphorous – $(NH_4)_2MoO_4$
Halogen – $AgNO_3$

68.



69. $N_1V_1 = N_2V_2$
$0.02 \times 1 \times V_1 = 0.01 \times 2 \times 10$, $V_1 = 10$ ml

70.



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71. $\text{RLVP}=\times \dfrac{P^0-P}{P^\circ}=\times \text{ solute}$

$\dfrac{1}{4}=\times \text{ solute}$

Let mass of urea = x

$$\frac{1}{4}=\frac{\frac{x}{60}}{\frac{x}{60}+\frac{1000}{18}}$$

$$\text{On solving } x=\frac{10000}{9}=1111.112$$

72. $\Delta G^\circ=\Delta H^\circ-T\Delta S^\circ$

$-RT\ln k=-54.07\times10^3-298\times10$

$-2.303\times8.314\times298\log K=-57050$

$$\log K=\frac{-57050}{-5705}=10$$

73. At a graph is decreasing ie Gibbs energy is decreasing ⇒ $\Delta G<0$

∴ Spontaneous at A

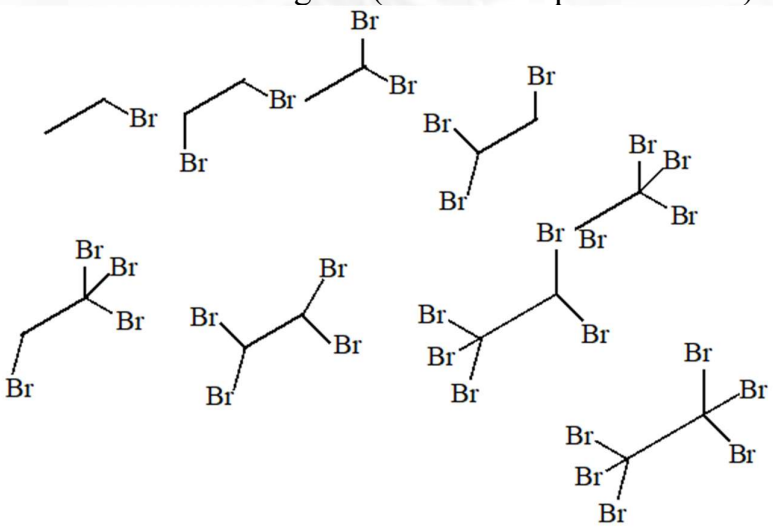
At b, equilibrium is attained

At c, graph is increasing

⇒$\Delta G>0$

∴ Non spontaneous at C.

74. SCN^- is ambidentat ligand (i.e. 2 nucleophilic centres)



75.

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